

SOC/SIEM Detection Report

Scheduled Task Persistence Detection

Wazuh + Sysmon + Windows Task Scheduler Operational Logs

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Report Type	Controlled SOC/SIEM Lab Incident Report
Scenario	Scheduled task persistence simulation
Date	May 14, 2026
Status	Resolved / Lab cleanup completed

Executive Summary

- A scheduled task named \LabPersistenceTest was created on WIN11-ENDPOINT to simulate a persistence technique in a controlled lab environment.
- Initial Wazuh searches did not show a clean scheduled-task registration event, which exposed a logging visibility gap.
- Local Sysmon evidence confirmed schtasks.exe activity. TaskScheduler Operational logging was then enabled and added to Wazuh agent log collection.
- After retesting, Wazuh received a TaskScheduler Event ID 106 event and mapped the activity to MITRE ATT&CK T1053.005 Scheduled Task.
- The test scheduled task was deleted and cleanup was verified.

Confidentiality Note: This report contains lab-only evidence. No production systems or real organizational data were used.

1. Environment

Component	Details
Host Platform	Windows 11 Pro with Microsoft Hyper-V
SIEM Server	Ubuntu Server running Wazuh-SIEM
Wazuh Manager IP	192.168.56.10
Endpoint	WIN11-ENDPOINT
Endpoint IP	192.168.56.20
Log Sources Used	Sysmon Operational, Windows TaskScheduler Operational, Wazuh Agent
Scenario Category	Persistence / Scheduled Task Creation

2. Key Findings

Finding Summary

- The scheduled task was successfully registered under the name \LabPersistenceTest.
- Local TaskScheduler Operational logging captured Event ID 106: task registered by WIN11-ENDPOINT\labuser.
- Wazuh ingested the TaskScheduler event from WIN11-ENDPOINT and generated a corresponding event record.
- The Wazuh event details showed MITRE mapping to T1053.005 Scheduled Task with tactics listed as Execution, Persistence, and Privilege Escalation.
- Cleanup was completed by deleting the scheduled task and confirming the task could no longer be queried.

3. Detection Logic and Visibility Improvement

Why this scenario matters: Scheduled tasks are commonly monitored in security operations because they can be used to launch commands automatically and maintain persistence on a Windows endpoint. In this controlled lab, the task action was harmless and only executed a whoami command that wrote output to a text file.

Visibility gap identified: The initial Wazuh event view did not produce a clear scheduled-task registration alert for the task name. The activity was then validated locally through Sysmon logs, confirming that endpoint telemetry existed. To improve detection coverage, the TaskScheduler Operational event channel was enabled and added to the Wazuh agent configuration.

```
wevtutil sl Microsoft-Windows-TaskScheduler/Operational /e:true

<localfile>
  <location>Microsoft-Windows-TaskScheduler/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

Restart-Service Wazuh
```

4. Activity Performed

The scheduled task was created using Administrator PowerShell on WIN11-ENDPOINT:

```
schtasks /create /tn "LabPersistenceTest" /tr "cmd.exe /c whoami > C:\Tools\LabPersistenceTest.txt" /sc once /st 23:59
```

The task was then queried and verified with:

```
schtasks /query /tn "LabPersistenceTest" /fo LIST /v
```

5. Evidence Summary

Evidence Source	Observed Evidence	Security Relevance
Local Sysmon Log	Process creation telemetry for schtasks.exe involving LabPersistenceTest.	Confirmed endpoint process-level telemetry existed.
Local TaskScheduler Operational Log	Event ID 106 recorded that WIN11-ENDPOINT\labuser registered task \LabPersistenceTest.	Confirmed the scheduled task registration at the Windows event log level.
Wazuh Threat Hunting	TaskScheduler event ingested from WIN11-ENDPOINT with Event ID 106 and taskName \LabPersistenceTest.	Confirmed SIEM visibility and successful log forwarding.
MITRE Mapping	Wazuh mapped the event to T1053.005 Scheduled Task.	Connects the behavior to a known persistence technique.
Cleanup Verification	schtasks /query returned an error showing LabPersistenceTest was no longer present.	Confirmed remediation and lab cleanup.

6. Wazuh Event Details

Field	Observed Value
Agent Name	WIN11-ENDPOINT
Agent IP	192.168.56.20
Log Channel	Microsoft-Windows-TaskScheduler/Operational
Windows Event ID	106
Task Name	\LabPersistenceTest
User Context	WIN11-ENDPOINT\labuser
Wazuh Rule ID	67014
Wazuh Rule Level	3
MITRE Technique	T1053.005 Scheduled Task
MITRE Tactics	Execution, Persistence, Privilege Escalation

7. Potential Impact

- A scheduled task can be used to run commands at a future time or repeatedly on a system.
- If unauthorized, scheduled task creation may indicate persistence, execution, or privilege escalation activity.
- A task running cmd.exe, PowerShell, scripts, or binaries should be reviewed carefully, especially when created by unexpected users or at unusual times.
- Monitoring TaskScheduler Operational logs provides better visibility than relying only on generic process creation events.

8. Investigation Steps

1. Created a scheduled task named \LabPersistenceTest using schtasks.exe on WIN11-ENDPOINT.
2. Attempted to locate the event in Wazuh using the task name and schtasks.exe-related searches.
3. Validated locally that Sysmon captured schtasks.exe activity, proving the endpoint generated useful telemetry.
4. Identified that the TaskScheduler Operational log was not providing usable event evidence at first.
5. Enabled the TaskScheduler Operational event channel using wevtutil.
6. Added Microsoft-Windows-TaskScheduler/Operational to the Wazuh agent configuration and restarted the Wazuh agent.
7. Recreated the scheduled task to generate fresh telemetry.
8. Confirmed local TaskScheduler Event ID 106 for task registration.
9. Confirmed Wazuh ingested the TaskScheduler event and mapped it to MITRE ATT&CK T1053.005.

9. Containment and Remediation

The test scheduled task was deleted using:

```
schtasks /delete /tn "LabPersistenceTest" /f
```

Cleanup was verified with:

```
schtasks /query /tn "LabPersistenceTest"
```

The query returned an error indicating the task could not be found, confirming the test artifact had been removed from the endpoint.

10. Lessons Learned

- Default SIEM visibility may not always surface the exact behavior being investigated; local validation is important.
- Sysmon is useful for process-level evidence, while TaskScheduler Operational logs provide clearer evidence for scheduled task registration.
- Adding targeted Windows event channels to Wazuh improves detection coverage and investigation quality.
- A strong SOC workflow includes generating activity, validating telemetry locally, confirming SIEM ingestion, and documenting remediation.

11. Skills Demonstrated

Skill Area	Evidence Demonstrated
SIEM Investigation	Searched and analyzed Wazuh Threat Hunting events for endpoint activity.
Windows Logging	Used TaskScheduler Operational and Sysmon logs to validate scheduled task activity.
Endpoint Telemetry	Confirmed schtasks.exe and task registration evidence on WIN11-ENDPOINT.
Detection Engineering Mindset	Identified a visibility gap, enabled a new log source, and retested.
MITRE ATT&CK Awareness	Recognized scheduled task activity as T1053.005-related behavior.
Remediation	Deleted the test task and verified cleanup.

Appendix A: Evidence Screenshots

The following screenshots document the scheduled task detection workflow and supporting evidence.

```
Administrator: Windows PowerShell
Id : 5
ProviderName : Microsoft-Windows-Sysmon
Message : Process terminated:
RuleName: -
UtcTime: 2026-05-14 22:46:47.305
ProcessGuid: {44f46d4c-50d7-6a06-8501-00000000a00}
ProcessId: 8596
Image: C:\Windows\System32\schtasks.exe
User: WIN11-ENDPOINT\labuser

TimeCreated : 5/14/2026 3:46:47 PM
Id : 1
ProviderName : Microsoft-Windows-Sysmon
Message : Process Create:
RuleName: -
UtcTime: 2026-05-14 22:46:47.280
ProcessGuid: {44f46d4c-50d7-6a06-8501-00000000a00}
ProcessId: 8596
Image: C:\Windows\System32\schtasks.exe
FileVersion: 10.0.26100.6725 (WinBuild.160101.0800)
Description: Task Scheduler Configuration Tool
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: schtasks.exe
CommandLine: "C:\WINDOWS\system32\schtasks.exe" /query /tn LabPersistenceTest /fo LIST /v
CurrentDirectory: C:\WINDOWS\system32\
User: WIN11-ENDPOINT\labuser
LogonGuid: {44f46d4c-4efd-6a06-913a-0e0000000000}
LogonId: 0xE3A91
TerminalSessionId: 2
IntegrityLevel: High
Hashes: SHA256-B96AF3066F089DE8A460F77BAAE006C5372A266BE18C561DAA7C7AE8635D399D
ParentProcessGuid: {44f46d4c-4f18-6a06-4901-00000000a00}
ParentProcessId: 7272
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
ParentUser: WIN11-ENDPOINT\labuser

TimeCreated : 5/14/2026 3:41:46 PM
Id : 5
ProviderName : Microsoft-Windows-Sysmon
Message : Process terminated:
RuleName: -
```

Figure 1. Local Sysmon evidence showing schtasks.exe process telemetry involving LabPersistenceTest.

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> Restart-Service Wazuh
PS C:\WINDOWS\system32> Get-Service *Wazuh*

Status      Name            DisplayName
-----
Running     WazuhSvc        Wazuh

PS C:\WINDOWS\system32> schtasks /delete /tn "LabPersistenceTest" /f
SUCCESS: The scheduled task "LabPersistenceTest" was successfully deleted.
PS C:\WINDOWS\system32> schtasks /create /tn "LabPersistenceTest" /tr "cmd.exe /c whoami > C:\Tools\LabPersistenceTest.txt" /sc once /st 23:59
SUCCESS: The scheduled task "LabPersistenceTest" has successfully been created.
PS C:\WINDOWS\system32> Get-WinEvent -LogName "Microsoft-Windows-TaskScheduler/Operational" -MaxEvents 50 | Where-Object {$_.Message -match "LabPersistenceTest"} | Select-Object TimeCreated, Id, ProviderName, Message -First 5 | Format-List

TimeCreated : 5/14/2026 5:01:26 PM
Id           : 106
ProviderName : Microsoft-Windows-TaskScheduler
Message      : User "WIN11-ENDPOINT\labuser" registered Task Scheduler task "\LabPersistenceTest"

TimeCreated : 5/14/2026 5:01:02 PM
Id           : 141
ProviderName : Microsoft-Windows-TaskScheduler
Message      : User "WIN11-ENDPOINT\labuser" deleted Task Scheduler task "\LabPersistenceTest"

PS C:\WINDOWS\system32> _
```

Figure 2. Local TaskScheduler Operational log showing Event ID 106 for task registration and Event ID 141 for task deletion.

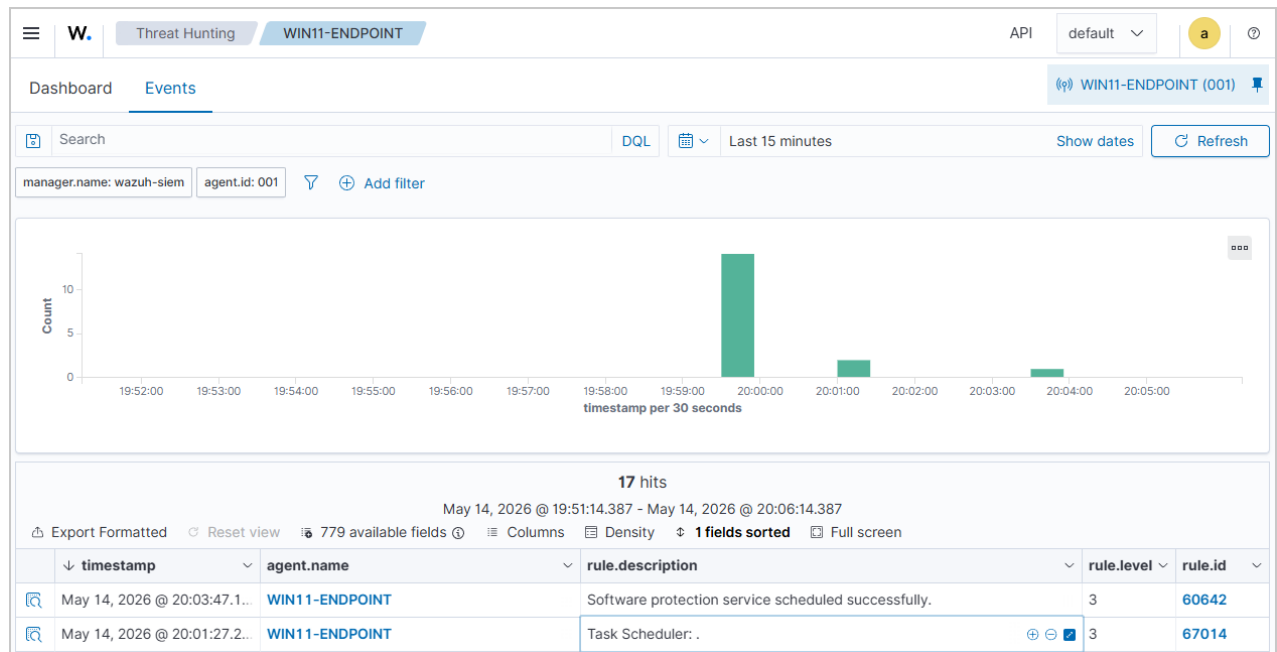


Figure 3. Wazuh Threat Hunting overview showing the Task Scheduler event from WIN11-ENDPOINT.

Document Details		view surrounding documents	view single document
t agent.ip	192.168.56.20		
t agent.name	WIN11-ENDPOINT		
data.win.eventdata.taskName	\\LabPersistenceTest		
data.win.eventdata.userContext	WIN11-ENDPOINT\\labuser		
t data.win.system.channel	Microsoft-Windows-TaskScheduler/Operational		
t data.win.system.computer	WIN11-ENDPOINT		
t data.win.system.eventID	106		
t data.win.system.eventRecordID	2		
t data.win.system.keywords	0x8000000000000000		
t data.win.system.level	4		
t data.win.system.message	"User "WIN11-ENDPOINT\\labuser" registered Task Scheduler task "\\LabPersistenceTest"		
t data.win.system.opcode	0		
t data.win.system.processID	1512		
t data.win.system.providerGuid	{de7b24ea-73c8-4a09-985d-5bdadcfa9017}		
t data.win.system.providerName	Microsoft-Windows-TaskScheduler		
t data.win.system.severityValue	INFORMATION		
t data.win.system.systemTime	2026-05-15T00:01:26.1155521Z		
t data.win.system.task	106		
t data.win.system.threadID	2228		
t data.win.system.version	0		
t decoder.name	windows_eventchannel		
t id	1778803287.1509		
t input.type	log		
t location	EventChannel		
t manager.name	wazuh-siem		
t rule.description	Task Scheduler: .		
# rule.firedtimes	1		
t rule.groups	windows, WEF		
t rule.id	67814		
# rule.level	3		
rule.mail	false		
t rule.mitre.id	T1053.005		
t rule.mitre.tactic	Execution, Persistence, Privilege Escalation		
t rule.mitre.technique	Scheduled Task		
timestamp	May 14, 2026 @ 20:01:27.200		

Figure 4. Wazuh event details showing TaskScheduler Operational Event ID 106, taskName \\LabPersistenceTest, and MITRE T1053.005 mapping.


```
Administrator: Windows PowerShell

-----
Running WazuhSvc          Wazuh
-----

PS C:\WINDOWS\system32> schtasks /delete /tn "LabPersistenceTest" /f
SUCCESS: The scheduled task "LabPersistenceTest" was successfully deleted.
PS C:\WINDOWS\system32> schtasks /create /tn "LabPersistenceTest" /tr "cmd.exe /c whoami > C:\Tools\LabPersistenceTest.txt" /sc once /st 23:59
SUCCESS: The scheduled task "LabPersistenceTest" has successfully been created.
PS C:\WINDOWS\system32> Get-WinEvent -LogName "Microsoft-Windows-TaskScheduler/Operational" -MaxEvents 50 | Where-Object {$_.Message -match "LabPersistenceTest"} | Select-Object TimeCreated, Id, ProviderName, Message -First 5 | Format-List

TimeCreated : 5/14/2026 5:01:26 PM
Id           : 106
ProviderName : Microsoft-Windows-TaskScheduler
Message      : User "WIN11-ENDPOINT\labuser" registered Task Scheduler task "\LabPersistenceTest"

TimeCreated : 5/14/2026 5:01:02 PM
Id           : 141
ProviderName : Microsoft-Windows-TaskScheduler
Message      : User "WIN11-ENDPOINT\labuser" deleted Task Scheduler task "\LabPersistenceTest"

PS C:\WINDOWS\system32> schtasks /delete /tn "LabPersistenceTest" /f
SUCCESS: The scheduled task "LabPersistenceTest" was successfully deleted.
PS C:\WINDOWS\system32> schtasks /query /tn "LabPersistenceTest"
ERROR: The system cannot find the file specified.
PS C:\WINDOWS\system32>
```

Figure 5. Cleanup verification showing LabPersistenceTest was deleted and could no longer be queried.